

CV

Jacinta D Kong

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Contact

[Website](#)

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Research and teaching appointments

- NSERC-Mitacs Alliance Postdoctoral Fellow
Department of Biology, Carleton University, Canada
1.2023 – Present
- Teaching and Research Fellow
Department of Zoology, School of Natural Sciences.
Trinity College Dublin, Dublin, Ireland
5.2019 – 1.2023
- Research Assistant
Intraspecific variation in mechanistic species distribution modelling,
The University of Melbourne, Australia
5.2018 – 12.2018
- Comparative Animal Physiology tutor (Second Year undergraduate)
School of BioSciences, the University of Melbourne, Australia
2017– 2018
- Ecology in Changing Environments tutor (Third Year undergraduate)
School of BioSciences, the University of Melbourne, Australia
2016– 2018
- Comparative Animal Physiology residential tutor,
University College, the University of Melbourne, Australia
2017– 2018
- First Year Chemistry residential tutor
St John's College, the University of Queensland Australia
8.2014 – 10.2014
- First Year Biology tutor (Science Learning Centre tutor),
the Faculty of Science, the University of Queensland, Australia
2012
- Peer Assisted Study Session leader: first year statistics
The Peer Assisted Study Session office and the School of Mathematics and Physics, the University of
Queensland Australia
8.2012 – 10.2012
- Peer Assisted Study Session leader: first year ecology
The Peer Assisted Study Session office, the University of Queensland Australia
2011 – 2012

- Volunteer laboratory technician. Animal husbandry.
White Evolutionary Physiology Laboratory, the University of Queensland
3.2012 – 10.2012
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Qualifications

Doctor of Philosophy (Science)

The University of Melbourne, Australia

Title: Predicting ectotherm life cycles under a variable climate: Physiological diversity of matchstick grasshopper eggs and their ecological and evolutionary implications

Repository access: <http://hdl.handle.net/11343/225704>

1.2015 – completed 14 Aug 2019, conferred 5.10.2020

Bachelor of Science (Honours Class I, University Medal)

The University of Queensland, Australia

Title: The effect of temperature on the relationship between metabolic rate and mass: Tests of the Metabolic Theory of Ecology

Conferred 6.12.2013

Bachelor of Science

The University of Queensland, Australia

Conferred 17.12.2012

Refereed journal articles

- Wu NC, Bovo RP, Enriquez-Urzelai U, Clusella-Trullas S, Kearney MR, Navas C, **Kong JD**: Global exposure risk of frogs to increasing environmental dryness. *Nature Climate Change* (2024), in press.
- Iosilevskii G⁺, **Kong JD**⁺, Meyer CG, Watanabe YY, Papastamatiou YP, Royer MA, Nakamura I, Sato K, Doyle TK, Harman L, Houghton JDR, Barnett A, Semmens JM, Maoiléidigh NÓ, Drumm A, O'Neill R, Coffey DM, Payne NL (2022) A general swimming response in exhausted obligate swimming fish. *Royal Society Open Science*. 9: 211869. DOI: [10.1098/rsos.211869](https://doi.org/10.1098/rsos.211869). ⁺Joint first author.
- Kearney MR, Jasper ME, White VL, Aitkenhead IJ, Blacket MJ, **Kong JD**, Chown SL, Hoffmann AA. (2022) Parthenogenesis without costs in a grasshopper with hybrid origins. *Science*. 376: 1110 – 1114 DOI: [10.1126/science.abm1072](https://doi.org/10.1126/science.abm1072). Altmetric score: 204
- Schwanz LE, Gunderson A, Iglesias-Carrasco M, Johnson MA, **Kong JD**, Riley J, Wu NC. (2022) Best practices for building and curating databases for comparative analyses. *Journal for Experimental Biology*. 225: jeb243295. DOI: [10.1242/jeb.243295](https://doi.org/10.1242/jeb.243295). Altmetric score: 12
- **Kong JD**, Hoffmann AA, Kearney MR. (2019) Linking thermal adaptation and life-history theory explains latitudinal patterns of voltinism. *Philosophical Transactions of the Royal Society B: Biological Sciences*. 374(1778). DOI: [10.1098/rstb.2018.0547](https://doi.org/10.1098/rstb.2018.0547). Altmetric score: 8
- Kearney MR, Deutscher J, **Kong JD**, Hoffmann AA. (2018) Summer egg diapause in a matchstick grasshopper synchronises the life cycle and buffers thermal extremes. *Integrative Zoology*. 13(4): 437–449. DOI: [10.1111/1749-4877.12314](https://doi.org/10.1111/1749-4877.12314). Altmetric score: 2

- Maino JL, **Kong JD**, Hoffmann AA, Barton MG, Kearney MR. (2016) Mechanistic models for predicting insect responses to climate change. *Current Opinion in Insect Science*. 17: 81 – 86. DOI: [10.1016/j.cois.2016.07.006](https://doi.org/10.1016/j.cois.2016.07.006). Altmetric score: 6
- **Kong JD**, Axford JK, Hoffmann AA, Kearney MR. (2016) Novel applications of thermocyclers for phenotyping invertebrate thermal responses. *Methods in Ecology and Evolution*. 7(10): 1201 – 1208. 2016. DOI: [10.1111/2041-210X.12589](https://doi.org/10.1111/2041-210X.12589). Altmetric score: 20

Preprints

- **Kong JD**, Ritchie MW, Vadboncoeur É, MacMillan HA, and Bertram SM. (2024) Growth, development, and life history of a mass-reared edible insect, *Gryllobates sigillatus* (Orthoptera: Gryllidae). bioRxiv. DOI: [10.1101/2024.10.08.617250](https://doi.org/10.1101/2024.10.08.617250).
- Muzzatti M, **Kong JD**, McColville E, Brzezinski H, Stabile C, MacMillan HA, Bertram SM: Larger diet particle sizes cause crickets to grow faster with no effect on final body size. bioRxiv. DOI: [2024.2024.2008.2013.607817](https://doi.org/2024.2024.2008.2013.607817).
- **Kong JD**, Wu NC. (2022) Can we improve our ability to identify climate vulnerability in ectotherm life cycles?. bioRxiv. DOI: [10.1101/2022.12.14.520433](https://doi.org/10.1101/2022.12.14.520433).
- **Kong JD**⁺, Arnoldi J-F⁺, Jackson AL, Bates AE, Morley SA, Smith JA, Payne NL⁺. (2022) Heating tolerance of ectotherms is explained by temperature's non-linear influence on biological rates. bioRxiv. DOI: [10.1101/2022.12.06.519315](https://doi.org/10.1101/2022.12.06.519315). ⁺Joint first author.

Research highlights

ORCID

[Google scholar](#)

H-index: 7 (GS)

Total citations: 178

- High attention papers in the top 5% of all research outputs scored by Altmetric:
 - **Science**: 22 News outlets, 1 blog, 86 tweeters. 1 Wikipedia page. Altmetric score: 223. Citations: 16
 - **Royal Society Open Science**: 90 tweeters. Altmetric score: 61. Citations: 14
- Ecology/evolution studies in some of the best journals in their respective disciplines, and in the top 25% of all research outputs scored by Altmetric:
 - **Methods in Ecology and Evolution**. Altmetric score: 20. Citations: 16
 - **Journal of Experimental Biology**. Altmetric score: 20. Citations: 16
 - **Philosophical Transactions of the Royal Society B**. Altmetric score: 8. Citations: 25
 - **Current Opinion in Insect Science**. Altmetric score: 6. Citations: 73

I have peer reviewed for:

- Global Change Biology
- Entomologia Experimentalis et Applicata
- Methods in Ecology and Evolution
- Nature Ecology and Evolution

- Ecological Entomology
 - The Canadian Entomologist
 - Environmental Entomology
 - Journal of Fish Biology
 - Current Zoology
 - Scientific Reports
 - Conservation Physiology
 - American Naturalist
 - Functional Ecology
 - Physiological Entomology
 - Journal of Evolutionary Biology
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Research grants or awards

TOTAL: \$13 750 AUD :australia: and **\$15 100 CAD** :canada:

- **\$7 500 CAD.** Summer 2024 NSERC-Mitacs Alliance Internship
- **\$100 CAD.** 4.2024 Canadian Society of Zoologists Travel Grant
- **\$7 500 CAD.** Summer 2023 NSERC-Mitacs Alliance Internship
- **\$2 500 AUD.** 6.2018 Holsworth Wildlife Research Endowment
Equity Trustees Charitable Foundation & the Ecological Society of Australia
- **\$300 AUD.** 12.2018 2nd runner up student presentation
Australian and New Zealand Society for Comparative Physiology and Biochemistry, Australia
- **\$500 AUD.** 8.2018 Student Travel Grant
To attend the 2018 Australian Entomological Society Annual Meeting, Australia
- **\$1 500 AUD.** 2018 Science Abroad Travelling Scholarship (SATS) Faculty of Science, the University of Melbourne, Australia
- **\$950 AUD.** 8.2018 FH Drummond Travel Award
School of BioSciences, the University of Melbourne, Australia
- **\$1 500 AUD.** 2016 School of BioSciences Research Higher Degree Financial Support for Research or Conference Travel
The University of Melbourne, Australia
- **\$6 000 AUD.** 2015 Holsworth Wildlife Research Endowment
Equity Trustees Charitable Foundation
- **\$500 AUD.** 2013 Three Minute Thesis (3MT) People's Choice Winner
The University of Queensland Undergraduate Research Conference

Other Awards and Scholarships

- 2024 Finalist. President's Award, Canadian Society of Zoologists Annual Meeting
- **\$13 541 AUD.** 2018 Research Training Program Scholarship, Australian Government
- **\$26 682 AUD.** 2017 Research Training Program Scholarship, Australian Government
- **\$200 AUD.** 2017 Runners-up in the Sustainability Prize photo competition, Graduate Student Association, the University of Melbourne
- **\$26 288 AUD.** 2016 Australian Postgraduate Award, Australian Government
- **\$25 849 AUD.** 2015 Australian Postgraduate Award, Australian Government
- 2014 University Medal 2013, the University of Queensland, Australia
- 2010 – 2013 Dean's Commendation for Academic Excellence (formerly Dean's Commendation for High Achievement), the Faculty of Science, the University of Queensland, Australia

Teaching contributions and course development

I have developed several graduate level tutorials or teaching material for journal or coding clubs that I provide freely through my website or through my GitHub.

2022 Trinity College Dublin implements a 4 year degree program with 2 years of general subjects (e.g. biological sciences stream, ~250 students) and 2 years towards a specific major (e.g. zoology, ~ 35 students). Total of 60 credits per year. Degree consists of mandatory core subjects and electives.

- Lecturer & course development: Statistics and computation for biologists (BYU22S01). 2nd year undergraduate core subject, 5 credits. Trinity College Dublin. Developed course lectures and practical material, implemented novel R packages for interactive teaching within the R environment (learnr package). In-person lectures & practicals.
- Lecturer & course development: Animal Diversity I (ZOU33003). 3rd year undergraduate, core zoology major subject, 5 credits. Trinity College Dublin. Developed course lectures and practical material. In-person lectures & practicals.
- Lecturer & course development: Animal Diversity II (ZOU33004). 3rd year undergraduate, core zoology major subject, 5 credits. Trinity College Dublin. Developed course lectures and practical material. In-person lectures & practicals.
- Contributing staff: Marine Biology (ZOU33000). 3rd year undergraduate, core zoology major subject, 5 credits. Trinity College Dublin. Resident field course based in Killary Fjord, Ireland.

2021

- Lecturer & course development: Statistics and computation for biologists (BYU22S01). 2nd year undergraduate core subject, 5 credits. Trinity College Dublin. Developed course lectures and practical material, implemented novel R packages for interactive teaching within the R environment (learnr package). Hybrid delivery: remote and in-person lectures & remote and in-person practicals.
- Lecturer & course development: Animal Diversity I (ZOU33003). 3rd year undergraduate, core zoology major subject, 5 credits. Trinity College Dublin. Developed course lectures and practical material. Hybrid delivery: remote lectures & in-person practicals.
- Lecturer & course development: Animal Diversity II (ZOU33004). 3rd year undergraduate, core zoology major subject, 5 credits. Trinity College Dublin. Developed course lectures and practical material. Hybrid delivery: remote lectures & in-person practicals.

2020

- Lecturer (Module co-ordinator) & course development: Statistics and computation for biologists (BYU22S01). 2nd year undergraduate. Trinity College Dublin. Developed course lectures and practical material, implemented reproducible workflow for online exams. Adapted for remote delivery.
- Lecturer & course development: Animal Diversity I (ZOU33003). 3rd year undergraduate. Trinity College Dublin. Developed course lectures and practical material. Module co-coordinator. Adapted for remote delivery.
- Lecturer & course development: Animal Diversity II (ZOU33004). 3rd year undergraduate. Trinity College Dublin. Developed course lectures and practical material. Adapted for remote delivery.

2019

- Lecturer & course development: Statistics and computation for biologists (BYU22S01). 2nd year undergraduate. Trinity College Dublin. Developed course lectures and practical material, implemented reproducible workflow for online exams.

- Lecturer & course development: Animal Diversity I (ZOU33003). 3rd year undergraduate. Trinity College Dublin. Developed course lectures and practical material. Acting module co-coordinator.
- Lecturer & course development: Animal Diversity II (ZOU33004). 3rd year undergraduate. Trinity College Dublin. Developed course lectures and practical material.

2017

- Guest lecturer: 2nd year Comparative Animal Physiology, University of Melbourne
- Course development: 2nd year Biostatistics, University of Melbourne. Evaluated course context and provided feedback.

2016

- Guest lecturer: 2nd year Comparative Animal Physiology, University of Melbourne

Conference presentations and invited talks

Presented by Jacinta unless otherwise stated.

2024

- The effects of diet and temperature on *Gryllobates sigillatus* life history. Presentation to Aspire Food Group, BugMars, and Western University. 23rd January 2024
- Leveraging physiology for insect mass rearing: effects of diet and temperature on cricket performance. Canadian Society of Zoologists Annual Meeting, Moncton, New Brunswick, Canada
- Grinding feed does not affect growth, development and survival in individually and group reared edible crickets. International Congress of Entomology, Kyoto, Japan. Poster
- Grinding feed does not affect growth, development and survival in individually and group reared edible crickets. Entomological Society of Canada Annual Meeting, Quebec City, Quebec, Canada.
- The effect of temperature on *Gryllobates sigillatus* life history. Presentation to Aspire Food Group. 26th August 2024
- The effect of laybox location on pinhead emergence. Presentation to Aspire Food Group and Cornell University. 16th October 2024

2023

- Leveraging physiology for insect mass rearing: effects of diet and temperature on cricket performance. Australian and New Zealand Society for Comparative Physiology and Biochemistry Conference, University of Queensland Gatton Campus, QLD, Australia

2022

- Heating tolerance of ectotherms is explained by temperature's non-linear influence on biological rates[^]. Society for Experimental Biology Annual Meeting, Montpellier, France. [^]presented by Nicholas Payne
- Can we improve our ability to understand ectotherm thermal tolerance? Society for Experimental Biology Animal Biology Early Career Researcher Symposium. 3-7th October. Tvärminne Zoological Station, Finland.
- Can we improve our ability to identify climate vulnerability in ectotherm life cycles? British Ecological Society Annual Meeting, Edinburgh, UK

2021

- Energetic turnover explains the inflexibility of upper thermal tolerances in ectotherms. Irish Ecological Association Meeting, University College Cork, Ireland (online)
- Ectotherm heat limits track biological rates. British Ecological Society Macroecology Special Interest Group meeting (online)
- Thermal adaptation and plasticity of egg development generates latitudinal patterns in insect life cycles under seasonal climates. Society for Experimental Biology Annual Meeting (online)

2019

- Detangling the complex problem of climate adaptation of insects living in a seasonal world. Victorian Biodiversity Conference, University of Melbourne, VIC, Australia
- Local adaptation of thermal responses generates voltinism patterns of matchstick grasshoppers, *Warramaba* (Orthoptera: Morabidae), along a latitudinal gradient. British Ecological Society Annual Meeting, Belfast, N. Ireland, UK

2018

- Selection against overwintering shapes thermal performance curves for development. Australian and New Zealand Society for Comparative Physiology and Biochemistry Conference, Monash University, VIC, Australia
- Environmental and developmental drivers at the egg stage generate divergent life cycles in wingless arid zone grasshoppers (Orthoptera: *Warramaba*). Australian Entomological Society Conference, Alice Springs, N.T., Australia
- The egg stage drives life cycle adaptation to climate in the widely distributed matchstick grasshoppers (*Vandiemena* and *Warramaba*, Orthoptera: Morabidae). ‘The height, breadth and depth of physiological diversity: variation across latitudinal, altitudinal and depth gradients’ Animal Biology Satellite Meeting, Florence, Italy, Society for Experimental Biology
- Microclimate-driven mechanistic models to examine clinal adaptation at the egg stage in a parthenogenetic grasshopper. Society for Experimental Biology Annual Conference, Florence, Italy

2017

- Does variation in egg developmental responses to temperature generate divergent life-cycles in a genus of flightless grasshoppers (*Warramaba* spp.)? School of BioSciences Postgraduate Symposium, the University of Melbourne, Parkville, Australia
- Egg development drives life cycles in *Warramaba* spp. grasshoppers. Australian and New Zealand Society for Comparative Physiology and Biochemistry Conference, Daintree Rainforest Observatory, QLD, Australia
- Mechanistic models for understanding and predicting insect responses to climate change. Australian Entomological Society Conference, Terrigal, N.S.W., Australia

2016

- Predicting insect egg development under variable climates. School of BioSciences Postgraduate Symposium, the University of Melbourne, Parkville, Australia
- Predicting egg development in the parthenogenetic grasshopper *Warramaba virgo* (Orthoptera: Morabidae). Australian and New Zealand Society for Comparative Physiology and Biochemistry Conference, Western Sydney University, N.S.W., Australia

2015

- Novel applications of thermocyclers for high-throughput phenotyping of invertebrate thermal response. Australian and New Zealand Society for Comparative Physiology and Biochemistry Conference, Fowler’s Gap, N.S.W., Australia

2013

- Every Breath You Take Links Metabolism and Ecology
- Three Minute Thesis, Undergraduate Research Conference, the University of Queensland, Australia
- The University of Queensland
- Every Breath You Take Links Metabolism and Ecology. Summer Research Introduction Session 2013, invited by the Office of Undergraduate Education, the University of Queensland, Australia
- Flying foxes and you: Exploring the exposure of society to so-called “rats with wings”
- Bachelor of Science Welcome Day, invited by the Faculty of Science, the University of Queensland, Australia

2012

- Flying foxes and you: Exploring the exposure of society to so-called “rats with wings”
- Advanced Study Program in Science Student Conference, the University of Queensland, Australia

2011

- Female-biased dispersal in the Eastern Water Dragon (*Physignathus lesueurii lesueurii*)
- Advanced Study Program in Science Student Conference, the University of Queensland, Australia

Professional service and affiliations

- 2023 – Present Member: Society for Comparative and Integrative Biology (SICB)
- 2024 – Present Member: Canadian Society of Zoologists (CSZ)
- 2015 – Present Member: Australian and New Zealand Society for Comparative Physiology and Biochemistry (ANZSCPB)
- 2019 – 2023 Member: British Ecological Society (BES) & Irish Ecological Association (IEA)
- 2018 Member: Royal Society of Victoria (RSV)
- 2017 – Present Member: Society for Experimental Biology (SEB)
- 2015 – 2020 Member: Australian Entomological Society (AES)
- 2019 – 2020 Member: European Society for Evolutionary Biology (ESEB)
- 2017 – 2018 President, BioSciences Postgraduate Society, the University of Melbourne
- 2016 – 2017 Vice President, BioSciences Postgraduate Society, the University of Melbourne
- Co-organiser of the Society for Experimental Biology Animal Biology Early Career Researcher Symposium. 3-7th October 2022. Tvärminne Zoological Station, Finland.
- Organiser of symposium “Leveraging insect physiology for mass rearing practice” for the 2024 International Congress of Entomology (ICE). 25-30 August 2024. Kyoto, Japan

- Co-organiser of symposium “Interdisciplinary advances in insect mass rearing in Canada” for the 2024 Entomological Society of Canada. 21-23 October 2024. Quebec City, Canada

I have mentored 3 undergraduate students, and 4 PhD students.

Professional development qualifications

- 14 Feb 2024 Mitacs Course: Applying the principles of sound leadership and team building
 - 29 Jan 2024 Mitacs Course: High performing leadership and teams
 - 26 Jan 2021 Epigeum Research Integrity Course
 - 19 Mar 2021 Dynamic Energy Budget Course (online)
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Community outreach and communication

2023

- Interview on CBC Radio about effects of hot weather on insects. 2nd October.

2022

- **Kong JD**, Chown SL, Hoffmann AA and Kearney MR. (2022) Reply to Adamo: No Signs of Pathogen Susceptibility in *Warramaba virgo*. Science. eLetter. DOI: [10.1126/science.abm1072](https://doi.org/10.1126/science.abm1072).
- Conference Organiser. Society for Experimental Biology Animal Biology Early Career Researcher Symposium. 3-7th October. Tvärminne Zoological Station, Finland.

2021

- Contributor to the Trinity Walton Club [STEM@Universi-TY](#) program, Trinity College Dublin (2021-present)
- Mentor for Irish Ecological Association mentoring meeting. 7th January
- The Socio-Economic Theory of Animal Abundance. April Fools blog post for [EcoEvo@TCD](#). 1st April

2020

- Profiled on Humans of BioSciences by the School of BioSciences, the University of Melbourne, Australia. [Website](#) & [Twitter](#). 17th December
- Guest interview with Newstalk radio, Ireland. 14th January

2019

- Mentor for BES Women in Science Mentoring Program
- Home and Away: 3 part blog series for [EcoEvo@TCD](#)
 - Home and Away: Would a Rosella by any other name smell as sweet (online 1 Nov)

- Home and Away: Monotreme mistakes (online 22 Nov)
- Home and Away: Australian expats (online 12 Dec)
- Victorian Biodiversity Conference Volunteer, Melbourne, Australia

2018

- Big Ideas in Macrophysiology. Report on the 2018 Animal Biology Satellite Meeting, Florence Italy. [Society for Experimental Biology Magazine](#)
- The University of Melbourne Open Day Volunteer, the University of Melbourne, Melbourne, Australia

2017

- The University of Melbourne Open Day Volunteer, the University of Melbourne, Melbourne, Australia
- Session Chair, BioSciences Postgrad Symposium, the University of Melbourne, Australia
- Victorian Biodiversity Conference Volunteer, Melbourne, Australia
- Blog post for Graduate Student Association, University of Melbourne on the biodiversity photo competition

2016

- Session Chair, BioSciences Postgrad Symposium, the University of Melbourne, Australia
- Novel applications of thermocyclers for characterising invertebrate thermal responses. [Video for Methods in Ecology and Evolution](#)

2015

- The University of Melbourne Open Day Volunteer, the University of Melbourne, Melbourne, Australia
- The Real Life of a Research Student. Science Undergraduate Research Journal (SURJ), Issue 2. The University of Queensland, Australia
- Jacinta on Zoology & Research. Interview with BITE BACK, Black Dog Institute, Australia. 31st August. [Link likely broken](#)

2014

- Brisbane Open House Volunteer, Brisbane Open House, Brisbane, Australia
- Student Chaperone for the International Baccalaureate World Student Conference, the University of Queensland, Australia
- Moreton Bay Research Station Open Day Assistant, Moreton Bay Research Station, Brisbane, Australia

2013

- Science Mentor to Science Undergraduate Students, appointed by the Faculty of Science, the University of Queensland, Australia

2012

- Moreton Bay Research Station Open Day Assistant, Moreton Bay Research Station, Brisbane, Australia